

PHY2049 - Lab 3: RC Circuit Capacitor Discharge

Date: June 14, 2025

Objective: measure the time constant of an RC circuit by recording capacitor voltage during discharge and fitting an exponential decay $V(t) = V_0 * \exp(-t / RC)$.

Results

Measured $\tau = 0.98 \text{ s}$ for $R = 10 \text{ k}\Omega$ and $C = 100 \text{ }\mu\text{F}$.
The exponential fit matches the data to within 2%.

Conclusion: the capacitor discharge follows the expected exponential law and the time constant agrees with theory.